

counter-surveillance techniques must be developed and deployed for all types of equipment. For existing material, applying radar cross-sectional reduction techniques could involve radical structural modification and, in some cases, may not be justifiable in terms of performance or cost. However, the selective use of thin radar absorbent coatings that apply to existing material would bring real benefits in terms of performance and cost-effectiveness under the right circumstances.

Radar is the most common method for detecting and tracking aircraft. Although radar is a necessary tool for air traffic management, the problem is in offensive military operations, which require the aircraft to attack its target unnoticed and escape. Radar involves the transmission of electromagnetic waves into the atmosphere; they then bounce off the aircraft onto the receiving antenna.

Conventional airplanes, such as airliners, are easily detected by radar due to their cylindrical shape in conjunction with engine knocks and the tailplane. The metals used in aircraft are also strong reflectors of electromagnetic waves and can be easily detected by radar. Compounds are also detected by radar, although usually not as easily as metals.

Radar absorbent material (RAM) is a specialised class of polymer based material applied to the surface of covert military aircraft, such as the F-22 Raptor and F-35 Lightning II, to reduce radar signature and thus complicate radar detection. These materials are also used in covert versions of unmanned tactical air systems such as the Boeing X-45. RAM is applied to the entire outer surface or (more often) to regions of high radar reflection, such as the edges of the surface.

RAM works on the principle that the aircraft absorbs energy from electromagnetic waves to minimise the intensity of the reflected signal. RAMs are used in combination with other stealth technologies, such as flat design and hidden engines, to make it difficult to detect military aircraft. It is possible to reduce the cross-section of a jet fighter's radar to the size of a medium-sized bird through ideal design and the application of stealth



*F-22 Raptor*

technologies. The information on the composition of the RAMs is kept secret by the military. Most RAMs consist of ferromagnetic particles embedded in a polymer matrix with a high dielectric constant.

One of the more common RAMs is called iron ball paint, which contains small metal coated spheres suspended in an epoxy based paint. The spheres are coated with ferrite or carbonyl iron. When electromagnetic radiation enters the iron ball paint, it is absorbed by the ferrite or carbonyl iron molecules, causing them to oscillate. Molecular oscillations then decay with the release of heat, which is an effective mechanism for damping electromagnetic waves. A small amount of heat generated by the oscillation is transferred to the aircraft structure where it is dissipated.

The second type of RAM consists of a neoprene foil containing ferrite particles or carbon black. This material, which was used in early versions of the F-117A Nighthawk, works on the same principle as iron ball paint by converting radar waves into heat. The USAF has introduced radar absorbent paints made of ferrofluidic and non-magnetic materials in some of its stealth aircraft. Ferrofluids are colloidal mixtures composed of ferromagnetic particles of nanometric size (less than 10nm) suspended in a carrier medium.

Ferrofluids are superparamagnetic, which means that they are strongly

polarised by electromagnetic radiation. When the fluid is subjected to an electromagnetic field strong enough, polarisation causes ripples to form on the surface. The electromagnetic energy used to form these ripples weakens or removes the energy of the reflected radar signal. RAM cannot absorb radar at all frequencies. The composition and morphology of the material is carefully adapted to absorb radar waves in a specific frequency band.

There are three main classes of polymers: thermosets, thermoplastics, and elastomers. The most widely used polymers in aircraft are thermosetting such as epoxy resins and bismaleimides, which are used as a matrix phase in fibre composites and as structural adhesives. The aerospace applications for thermoplastics and elastomers are much smaller.

The properties of polymers that make them useful aerospace materials are their low weight, excellent corrosion resistance, and good ductility. Some polymers are also transparent for use in window applications. However, polymers have low stiffness, strength, fatigue life, and creep resistance, and therefore should not be used alone in structural applications.

Polymers are produced by polymerisation reactions called addition or condensation reactions, which link monomer units in macromolecules in the form of long chains. The chemical and physical structure of the molecu-